	National University of Computer and Emerging Sciences Lahore Campus **COAL (Computer Organization and Assembly Language) — EE2003**	**Quiz # 2** **BCS -3H , BDS-3A** **CLO # 2** Date: _____ Total Marks: 10

Course Instructor: Sana Fatima		
Name:	**Roll No:**	**Section:**

Important Instructions:

- *In case of plagiarism, all quizzes will be marked as zero.*
- *Mention your roll number, name and section properly while submitting.*
- *Show proper working for full credit.*

Q1: Write a subroutine that receives a parameter (memory address) and performs 32 bit shift right on that memory location.

Solution:

```
[org 0x0100]
jmp start

     num: dd 0x12345678

shiftRight32:
     push bp
     mov  bp, sp
     push bx

     mov  bx, [bp+4]

     shr  word [bx+2], 1
     rcr  word [bx], 1

     pop  bx
     pop  bp
     ret  2

start:
     mov  ax, num
     push ax
     call shiftRight32

mov  ax, 0x4c00
int  0x21
```